

unpredictably, and once it has been randomised, the superposition has quietly degraded into an ordinary classical probability distribution. T2 is also typically in the range of 100 to 300 microseconds on Heron-class devices. Since a single-qubit gate takes only a few tens of nanoseconds, we have room for a few thousand operations before decoherence destroys the computation—which sounds generous until you recall that a useful quantum algorithm may need millions of operations.

Gate errors arise because the microwave pulses that implement our gates are not perfect. The pulse may be slightly too long, slightly off frequency, or slightly the wrong shape, so the state vector does not rotate through exactly the intended angle. On present-day Heron processors, single-qubit gates are extraordinarily good, with error rates of the order of a few parts in ten thousand. Two-qubit gates are roughly ten times worse, with error rates of a few parts in a thousand. That ten-to-one ratio shapes the whole of practical quantum programming. In the NISQ era, the number of two-qubit gates is the currency in which we pay for our algorithms, and circuit designers count them the way a jeweller counts grams. Our Bell circuit spends exactly one, which is why it came back so nearly intact.

Readout errors, also called measurement errors, occur at the very last step. The state of a superconducting qubit is read by bouncing a microwave signal off a resonator coupled to it and inspecting how the signal comes back. Occasionally the classification is simply wrong, and a qubit in $|0\rangle$ is reported as 1. Readout is usually the noisiest single operation on the chip, with error rates around one percent or a little below. Together with imperfect initialisation, these are known as SPAM errors—state preparation and measurement errors.

Crosstalk is the failure of qubits to mind their own business. Driving one qubit can disturb its neighbours, and two qubits that are physically adjacent can retain a small unwanted interaction even when we are not using it. This is precisely the problem that the tunable couplers of the Heron architecture were designed to solve: when a coupler is not needed, it is detuned so that the residual interaction is suppressed.

Leakage is a reminder that our tidy two-level picture is a convenient fiction. A transmon qubit (a type of superconducting charge qubit) is really an oscillator with many energy levels, of which we have borrowed the lowest two and named them $|0\rangle$ and $|1\rangle$. A badly shaped pulse can push the device into the second excited state, outside the computational space altogether. Such a qubit has not merely made an error—it has left the game.

Finally, there are the material and environmental villains. Microscopic defects in the oxide layers of the chip, known as two-level systems or TLS defects, drift in and out of resonance with the qubits and steal energy from them; the Heron R2 generation includes dedicated TLS mitigation for exactly this reason. And, spectacularly, cosmic rays occasionally strike the substrate and produce bursts of

correlated errors across many qubits at once. It is a humbling thought that a quantum processor in a laboratory in the United States can be disturbed by a particle that began its journey outside the solar system

At this point, it is worth revealing something that the three-level approach of this series obliges us to confront. The CNOT gate we wrote in *qiskit10.py* does not exist on the chip. The Heron processor implements only a small set of native operations: the single-qubit gates X, SX (the square root of X) and RZ, and the two-qubit CZ (controlled-Z) gate. When *transpile()* prepares our circuit for the backend, our elegant Hadamard-plus-CNOT pair is rewritten as a sequence of RZ and SX rotations surrounding a single CZ gate. Try adding the statement *print(transpiled_circuit.draw())* to the program and see for yourself. The result is shown in Figure 2, and every gate in it is drawn from that short native list. Every physical error we have just discussed attaches itself to those operations, not to the idealised gates we wrote. Figure 2 also shows something the drawing of the original circuit could not: the labels *q_0 -> 22* and *q_1 -> 23* record which physical qubits on the 156-qubit chip the transpiler chose for us. Here we were fortunate, because qubits 22 and 23 are already neighbours on the lattice. Had they not been, the transpiler would have had to insert SWAP operations to bring them together, and each SWAP costs three more CZ gates.

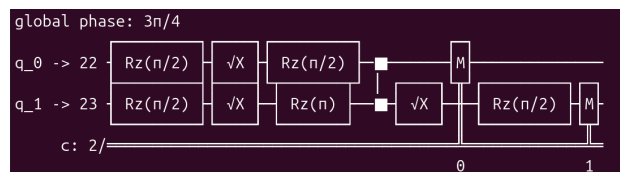


Figure 2: The Bell circuit of *qiskit10.py* after transpilation for *ibm_kingston*

The four Bell states

We are now ready for the central idea of this article. In 1935, Albert Einstein, Boris Podolsky and Nathan Rosen argued that quantum mechanics might not provide a complete description of reality. Their argument concerned entangled correlations: measuring one particle could reveal something about a distant partner. Surely, they reasoned, the result must already have been encoded in the pair through hidden variables that quantum mechanics had omitted. For nearly thirty years, the Einstein-Podolsky-Rosen (EPR) paradox remained largely a conceptual and philosophical dispute. That changed in 1964, when the Northern Irish physicist John Stewart Bell showed that local hidden-variable theories and quantum mechanics make measurably different predictions. A philosophical argument had become an experiment. Alain Aspect, John Clauser, and Anton Zeilinger conducted increasingly sophisticated experiments that supported the quantum predictions, and in 2022 they shared the Nobel Prize in Physics. Regular readers of this series may by now expect